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FEDERAL SPECIFICATION

SEALER, WATER AND WEATHER RESISTANT, FOR ASPHALT, CONCRETE, AND MASONRY SURFACES

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the properties of a sealer to impart water and weather resistance to asphalt, concrete, and masonry surfaces.

1.2 Classification. The sealer shall be furnished in the following types, classes and grades as specified (see 6.3).

Type I - For asphalt surfaces.
  Class A - Horizontal surfaces.
    Grade 1 - Gloss.
    Grade 2 - Flat.

Type II - For concrete and open-pore masonry.
  Class A - Horizontal surfaces.
    Grade 1 - Gloss.
    Grade 2 - Flat.
  Class B - Vertical surfaces.
    Grade 1 - Gloss.
    Grade 2 - Flat.

Type III - For closed-pore masonry.
  Class B - Vertical surfaces.
    Grade 1 - Gloss.
    Grade 2 - Flat.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-P-19	-	Paint, Acrylic Emulsion, Exterior.
TT-P-85	-	Paint, Traffic, Reflectorized, for Airfield Runway Marking (Drop-on Type).
VV-G-632	-	Grease, Industrial, General Purpose.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter.
PPP-D-729	-	Drums, Shipping and Storage, Steel, 55-Gallon.
PPP-P-704	-	Pails, Metal: Shipping, Steel (1 through 12 Gallon).
PPP-P-1892	-	Paint, Varnish, Lacquer and Related Materials; Packaging, Packing and Marking Of.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

FSC 8030

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

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Military Specification:

MIL-L-60326 - Lubricant, Fluorocarbon Telomer Dispersion (For Use with Ammunition).

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

County of Los Angeles Air Pollution Control Regulations:

Rule 66 - Organic Solvents.

(Application for copies should be addressed to the Air Pollution Control District, County of Los Angeles, 434 South San Pedro Avenue, Los Angeles, CA 90013.)

American Society for Testing and Materials (ASTM) Standards:

D 56 - Flash Point by Tag Closed Tester.
D 543 - Resistance of Plastics to Chemical Reagents.
D 609 - Preparation of Steel Panels for Testing Paints, Varnish, Lacquer and Related Products.
D 882 - Tensile Properties of Thin Plastic Sheeting.
D 1200 - Viscosity of Paints, Varnishes, and Lacquers by Ford Viscosity Cup.
D 1308 - Effect of Household Chemicals on Clear and Pigmented Organic Finishes.
D 1653 - Moisture Vapor Permeability of Organic Coating Films.
D 2369 - Volatile Content of Paints.
D 2620 - Light Stability of Clear Coatings.

(Copies of the above publications may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

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Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. The sealer shall be furnished as a single component clear material that shall not discolor when exposed to ultraviolet radiation, when tested in accordance with table III. The component, when completely cured, shall maintain a seal against moisture, spalling, liming, and dusting action. The sealer shall be compatible with dry paint films.

3.2 Solvent. The volatile portion of the sealer shall be of such composition as to render the compound suitable for use under Air Pollution Regulations, when tested in accordance with table III.

3.3 Storage stability. The sealer when stored for one year at -18 to + 46°C (0 to 115°F), and tested in accordance with table III, shall meet all the requirements of the specification for applicable type, grade and class.

3.4 Working properties. The sealer, at material temperatures ranging from 10 to 32°C (50 to 90°F), shall be capable of uniform application with a suitable brush, roller, or spray at a covering rate of approximately 200 square feet per gallon, when tested in accordance with table III.

3.5 Flexibility. The dried film shall show no evidence of cracking or chipping when tested in accordance with 4.4.5.

3.6 Adhesion. The material shall have good adhesion to the substrate when tested in accordance with 4.4.6.

3.7 Corrosion resistance. The material shall not be affected by a 10 percent solution of calcium chloride, when exposed for 500 hours as specified in 4.4.7.

3.8 Weathering. After 600 hours of weathering, the material shall show no blistering, cracking, crazing, or yellowing, when tested in accordance with 4.4.4.

3.9 Condition in container. The material shall show no evidence of separation. Any settling of pigment shall be soft. Any separation or settling shall be readily broken up with a paddle and easily redispersed to its proper condition by stirring. There shall be no putrefaction, livering or gelling.

3.10 Quantitative requirements. The compound shall meet the requirements as specified in table I, when tested as specified in table III.

3.11 Resistance to grease and various liquids. The material shall meet the following requirements as specified in table II, when tested in accordance with 4.4.10.

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TABLE I

Characteristics	Type I Class A Grade 1 Min.	Type I Class A Grade 1 Max.	Type I Class A Grade 2 Min.	Type I Class A Grade 2 Max.	Type II Class A Grade 1 Min.	Type II Class A Grade 1 Max.	Type II Class A Grade 2 Min.	Type II Class A Grade 2 Max.	Type II Class B Grade 1 Min.	Type II Class B Grade 1 Max.	Type II Class B Grade 2 Min.	Type II Class B Grade 2 Max.	Type III Class B Grade 1 Min.	Type III Class B Grade 1 Max.	Type III Class B Grade 2 Min.	Type III Class B Grade 2 Max.
Total solids (percent by weight).	35	-	35	-	35	-	35	-	28	-	28	-	12	-	12	-
Viscosity (No. 4 Ford cup, sec.)	-	68	-	68	-	68	-	68	-	45	-	45	-	30	-	30
Flashpoint (°C)	38	-	38	-	38	-	38	-	38	-	38	-	38	-	38	-
(°F)	100	-	100	-	100	-	100	-	100	-	100	-	100	-	100	-
Drying time:																
Set-to-touch (min.)	-	15	-	15	-	30	-	30	-	30	-	30	-	30	-	30
Dry-hard (hrs.)	-	24	-	24	-	48	-	48	-	48	-	48	-	48	-	48
Moisture vapor transmission Rate (dry film-gms/sq. centimeter/24 hours	-	0.01	-	0.01	-	0.01	-	0.01	-	0.006	-	0.006	-	0.006	-	0.006
Abrasion resistance:																
Wear index (mg. wt. loss/-500 cycles)	-	60	-	60	-	40	-	40	-	-	-	-	-	-	-	-
Wear cycles (to break through at 200 sq. ft./gal. spread rate	300	-	300	-	500	-	500	-	-	-	-	-	-	-	-	-
Elongation (percent)	50	-	50	-	25	-	25	-	50	-	50	-	50	-	50	-
Gloss (60° percent)	75	-	-	10	75	-	-	10	75	-	-	10	75	-	-	10

SS-S-1996

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TABLE II

Type	Grease 1/	SAE 30 Oil	100 Octane Gasoline	JP4 Jet Fuel
Types I, II and III initial (after 2 hours)	I 2/	I	T 3/	T
after 24 hours	I	I	I	I

1/ Conforming to VV-G-632.
2/ I-Film intact, no residual discoloration.
3/ T-Film tacky but not dissolved.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling and inspection shall be in accordance with method 1031 of Fed. Test Method Std. No. 141.

4.3 Inspection of preparation for delivery requirements. Packaging, packing and marking shall be inspected in accordance with PPP-P-1892.

4.4 Test procedures.

4.4.1 The material shall be tested in accordance with the following applicable methods of Fed. Test Method Std. No. 141, except as otherwise specified in table III. Alternate test techniques and equipment may be used by the Government, but in case of dispute, the methods specified here shall prevail. Unless otherwise specified, all test specimens shall be prepared and tested at a temperature of 23 ± 1°C (73 ± 2°F) and a relative humidity of 50 ± 5 percent. Failure to pass any test or noncompliance to the requirements shall be cause for rejection of the lot. All tests shall be evaluatedfor conformance to the requirements specified in section 3.

TABLE III

Characteristics	Applicable Test Method: Fed. Test Method Std. No. 141	Applicable Test Method: ASTM Method	Applicable Test Method: Paragraph Reference	Requirement Reference
Total solids (% by weight)	---	D 2369	---	Table I
Viscosity	---	D 1200	---	Table I
Flashpoint	---	D 56	---	Table I
Drying time	4061	-----	4.4.2	Table I
Moisture vapor transmission rate	---	D 1653	4.4.3	Table I
Weathering resistance	6152	-----	4.4.4	3.8
Flexibility	6221	-----	4.4.5	3.5
Adhesion	6301	-----	4.4.6	3.6
Corrosion resistance	---	D 543	4.4.7	3.7
Abrasion resistance (Taber)	6192	-----	4.4.8	Table I
Elongation	---	D 882 Method A	4.4.9	Table I
Storage stability	3022	-----	---	3.3
Condition in container	3011	-----	---	3.9
Gloss	6101	------	---	Table I
Resistance to grease and various liquids	---	D 1308	4.4.10	Table II
Solvents (volatile matter)	4351	-----	---	3.2
Discoloration (light stability)	---	D 2620	---	3.1
Working properties	4541	-----	---	3.9

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4.4.2 Drying time. Apply a coat of the material which results in a 0.025 ± 0.003 mm (0.001 ± 0.0001 inch) dry-film thickness on a clean glass panel and allow to dry in a horizontal position at standard conditions. Testing and examination shall be as specified in table III.

4.4.3 Moisture vapor transmission rate. The test as specified in table III shall be conducted on a 0.051 ± 0.005 mm (0.002 ± 0.0002 inch) unsupported dry film, prepared at standard conditions as specified in 4.4.9.

4.4.4 Weathering resistance. Panel preparation shall conform to Procedure A in method 2011 of Fed. Test Method Std. No. 141. Dip the panel in the sealer, and allow to air dry for a minimum of 72 hours. Testing shall be as specified in table III.

4.4.5 Flexibility. Panel preparation, for tests as specified in table III, shall be the same as in 4.4.4. The sealer shall be applied by draw down to a prepared panel conforming to ASTM D 609, Type 3B, to give a dry film thickness of 0.025 ± 0.003 mm (0.001 ± 0.0001 inch). Allow the coated panel to dry in a horizontal position at standard conditions for 72 hours. Bend double rapidly over a 3.18 mm (1/8 inch) mandrel, and examine without magnification.

4.4.6 Adhesion. The tests, as specified in table III, shall be conducted on coated and uncoated concrete and asphalt blocks with and without water immersion. The total number of tests to be performed is eight; two blocks of each material are sufficient.

4.4.6.1 Preparation of blocks.

4.4.6.1.1 Concrete block preparation. A ready-mixed composition consisting of Portland cement, washed sand, and washed gravel 1/ shall be used. A mold with a removable base plate shall be constructed of wood or metal, and treated with a release agent such as water-white mineral oil. The cement-aggregate composition shall be mixed to uniformity in the ratio of 3.5 quarts clean water to 80 pounds composition. The mold shall be filled with the above mixture to overflowing. This mass shall then be screeded (leveled) to a sand paper like surface with a wooden float, and finally leveled off with a metal straight edge drawn across the top with a sawing motion. The blocks shall be cured under burlap, or similar material, which is kept soaked with water, for four days. After cure, the blocks shall air dry for 48 hours prior to use.

4.4.6.1.2 Asphalt block preparation. A ready-mixed composition consisting of an asphalt cut-back, sand, and gravel 2/ shall be used. A mold shall be constructed of wood. The ready mixed composition shall be maintained at 49°C (120°F) for 24 hours, and then poured into the mold and compacted by tamping with a fifty-pound weight. Allow a weight of 10 pounds per square inch, covering the block surfaces, to remain for 48 hours. Remove weights and air dry the blocks for 48 hours prior to use.

4.4.6.2 Coating of blocks.

4.4.6.2.1 Concrete blocks, prepared as specified in 4.4.6.1.1, shall be coated by applying to half of the block TT-P-19, at a spread rate of about 300 square feet per gallon, and to the other half TT-P-85 (white) at a spread rate of about 100 square feet per gallon, allowing this coated block to dry for 24 hours at standard conditions before applying the sealer.

4.4.6.2.2 Asphalt blocks, prepared as specified in 4.4.6.1.2, shall be coated as specified in 4.4.6.2.1.

4.4.6.3 Sealing of blocks. The sealer shall be applied by brush, roller or spray at a spreading rate of about 200 square feet per gallon to substrates specified in 4.4.6.1.1, 4.4.6.1.2, 4.4.6.2.1, and 4.4.6.2.2, and allowed to dry for 72 hours at standard conditions prior to testing.

1/ Sacrete Concrete Mix from Campbell Products, Towson, Maryland 21204, is satisfactory.

2/ Sacrete Black Top from Campbell Products, Towson, Maryland 21204, is satisfactory.

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4.4.6.4 Performance of adhesion tests.

4.4.6.4.1 Uncoated and coated concrete and asphalt blocks with the sealer applied and not immersed in water shall be subjected to the tape test indicated in Method 6301 of Fed. Test Method Std. No. 141, and evaluated for conformance with 3.6.

4.4.6.4.2 Uncoated and coated concrete and asphalt blocks, with the sealer applied, shall be tested in accordance with Method 6301 of Fed. Test Method Std. No. 141. After submersion of the blocks for 24 hours in distilled water, they shall be removed from the water and tilted to approximately 60 degrees from the vertical and allowed to dry for two hours prior to evaluation for conformance with 3.6.

4.4.7 Corrosion resistance. The test shall be in accordance with Procedure I of the method specified in table III. The sealer shall be applied by draw down to a substrate described in 4.4.9 so as to produce an unsupported dry film having a thickness of 0.025 ± 0.003 mm (0.001 ± 0.0001 inch). The coated panel shall dry in a horizontal position for 24 hours at standard conditions prior to testing.

4.4.8 Abrasion resistance. The wear index (loss in weight) shall be determined for 500 cycles, and the number of wear cycles to break through shall be determined using CS 17 wheels with a 500-gram head load. Testing shall conform as specified in table III. The sealer shall be applied by brush, roller, or spray to a substrate suitable for the test at a covering rate of approximately 200 square feet per gallon, and dried at standard conditions for 24 hours for Type I material, 48 hours for Type II material, and 48 hours for Type III material.

4.4.9 Elongation. A smooth, flat, polished glass plate, having nominal dimensions of 200 x 300 mm (8 x 12 inches), shall be uniformly covered with a sheet of hexafluoropropylene 3/, which is then uniformly coated with a material conforming to MIL-L-60326, Type I 4/, and allowed to dry in a horizontal position for 24 hours at standard conditions. Place a draw down blade over the coated hexafluoropropylene-glass plate combination and apply the sealer. The gap clearance of the draw down blade shall be such that the resultant dry film thickness is 0.051 ± 0.005 mm (0.002 ± 0.0002 inch). Allow to air dry for 96 hours in a horizontal position at standard conditions. After air drying, carefully remove the film from the substrate and cut from it three specimen strips 1 x 4 inches each. The cut strips are tested as specified in table III at a separation rate of two inches per minute with an initial jaw separation of two inches.

4.4.10 Resistance to grease and various liquids. Apply a coat of the material which results in a 0.025 ± 0.003 mm (0.001 ± 0.0001 inch) dry film thickness, run on a 32 gauge steel panel, of nominal dimensions of 10 by 15 cm (4 by 6 inches), prepared in accordance with method 2011 of Fed. Std. No. 141. Allow to air dry for 72 hours in a horizontal position at standard conditions. Subject panels to test media cited in table II in accordance with table III for six hours, wipe clean with a soft wiping cloth, allow to stand for two hours at standard conditions and test for compliance with table II. The grease shall be applied by spreading approximately 10 grams on the dried film over an area of approximately 50 x 50 mm (2 x 2 inches) and following the same procedure indicated above.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.3).

5.1.1 Level A. The sealer shall be furnished in 1-gallon cans conforming with PPP-C-96, type V, class 2; 5-gallon pails conforming with PPP-P-704, type I class 2, 3, 4, 5, or 8; or in 55-gallon drums conforming with PPP-D-729, type I or II (see 6.3).

3/ Teflon FEP 2 mil film thickness (Card No. 03111, item No. 29499) from E.I. DuPont de Nemours Co., Inc., Wilmington, Delaware 19898 is satisfactory.

4/ Dry lubricant (MS-122 Fluorocarbon Release Agent) from Miller-Stephenson Chemical Co., Danbury, Connecticut 06810 is satisfactory.

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5.1.2 Level B. Packaging shall comply with paragraph 5.1.1, except commercial coatings are acceptable (see 6.3).

5.1.3 Level C. The sealant shall be furnished in 1-gallon, 5-gallon, or 55-gallon quantities as specified (see 6.3), in containers furnished for commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.3).

5.2.1 Level A. The filled 1-gallon cans shall be packed in quantities of 4 in a close-fitting box conforming with PPP-B-636, class weather resistant, V3c minimum. Closure shall be in accordance with the appendix of PPP-B-636. The 5- and 55-gallon containers require no overpacking.

5.2.2 Level B. The level B packing shall comply with paragraph 5.2.1 except that the box shall comply with PPP-B-636, class domestic.

5.2.3 Level C. The sealer shall be packed in a manner which will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier. Containers and packing shall comply with Uniform Freight Classification or National Motor Freight Classification. The compound shall be furnished in 1-gallon cans, 5-gallon pails, or 55-gallon drums.

5.3 Marking.

5.3.1 Special labeling. Packages shall be labeled to comply with the requirements of Federal laws and regulations in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. The sealer is intended for either roller, spray, or brush application to finished asphalt surfaces and to interior or exterior finished concrete, brick, or masonry surfaces, producing a waterproof coating that prevents weathering, spalling, liming, and dusting action. The compound shall retard surface staining and erosion, and also retard formation of efflorescent salts on brick and other masonry surfaces. Use of the sealer at air fields is limited to parking aprons. The sealer should be applied at temperatures ranging from 10 to 32°C (50 to 90°F) with the relative humidity not in excess of 55 percent. Old surfaces shall be clean and free of dirt, grease, oil and other foreign matters. The user is cautioned not to add any material to the ready-for-use sealer which might adversely affect its performance.

6.2 Application. The sealer shall be applied to dry, clean asphalt, concrete, and masonry surfaces. Full cure is normally reached at least 72 hours after the removal of such wet films as burlap or polyethylene from the surfaces.

6.3 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class and grade required (see 1.2).
(c) Selection of applicable level of packaging and packing required (see 5.1 and 5.2).
(d) Size of container required (see 5.1.1, 5.1.2, and 5.1.3).

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